Chapter 11

Advanced Topics in Hypothesis Testing

This chapter covers some additional hypothesis tests besides the methods covered in the previous chapter. Parametric hypothesis tests, e.g., t-test, rely on several key assumptions such as that (1) the data are drawn from normally distributed populations, (2) variances are equal across groups, and (3) observations are independent. In real-world applications, these assumptions are often violated. Economic data like household income, reaction times in psychology, or hospital stays in health research tend to be skewed, contain outliers, or have unequal variances between groups. Even when data are measured on ordinal or discrete scales—such as survey responses or counts—the continuity and normality assumptions of parametric tests are not satisfied. Violations of independence are also common, particularly in repeated-measures designs or clustered data, leading to misleading inferences if traditional methods are applied.

One might expect the Central Limit Theorem (CLT) to overcome many of these issues by ensuring that, with sufficient sample size, the sampling distribution of the mean becomes approximately normal. However, the CLT's guarantees hold only under specific conditions. More specifically, it assume that the data is independent, identically distributed, and possess a finite variance. In small samples or with strongly skewed or heavy-tailed distributions, the normal approximation can remain poor even with dozens of observations. Likewise, dependence among observations or infinite variance—common in financial or environmental data—violates the conditions of the theorem, leaving parametric tests unreliable.

These practical limitations motivate the use of non-parametric—also known as distribution-free—methods. Non-parametric tests make fewer assumptions about the underlying population and are often based on ranks or signs rather than raw values, providing robustness against skewness, outliers, and non-normality. They allow researchers to test hypotheses about medians, distributions, or associations

without requiring strong parametric conditions, making them essential tools when data deviate from idealized statistical assumptions.

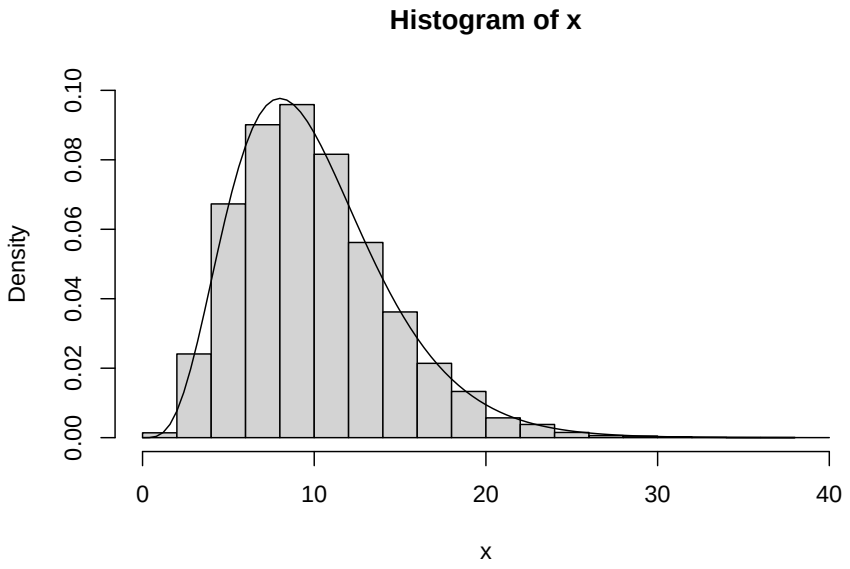
11.1 Chi-Square Test (χ^2-Test)

The χ^2-test is used to conduct a hypothesis test on qualitative variables. Before introducing the procedure, a presentation of the continuous χ^2-distribution is necessary.

Given values $x_1, x_2, \ldots, x_k$ independently drawn from standard normal distribution, then the squared sum of those values follows a χ^2-distribution for which k specifies the degrees of freedom:

$$\sum_{i=1}^{k} x_i^2 = X \sim \chi_k^2$$

Consider the following simulation for which $k = 10$. We randomly generate 10 outcomes from a standard normal distribution and then calculate the sun of the squared outcomes. Given 10,000 draws of the 10 outcomes from a standard normal distribution, the following histogram is obtained.



To conduct a chi-square hypothesis test, consider the data on voting and education in `gss`. Using the function `CrossTable` associated with the package gmodels, the following table can be constructed for the survey year 2022.

```
##
```

```
##
##    Cell Contents
## |-------------------------|
## |                       N |
## | Chi-square contribution |
## |-------------------------|
##
##
## Total Observations in Table:  3155
##
##
##                    | df$vote20
##      df$schooling |         0 |         1 | Row Total |
## -------------------|-----------|-----------|-----------|
##      Less than HS |       132 |       134 |       266 |
##                   |    96.887 |    26.669 |           |
## -------------------|-----------|-----------|-----------|
##       High School |       399 |      1054 |      1453 |
##                   |    23.240 |     6.397 |           |
## -------------------|-----------|-----------|-----------|
## Asso./Junior Col. |        47 |       246 |       293 |
##                   |     4.172 |     1.148 |           |
## -------------------|-----------|-----------|-----------|
##          Bachelor |        63 |       635 |       698 |
##                   |    51.006 |    14.040 |           |
## -------------------|-----------|-----------|-----------|
##          Graduate |        40 |       405 |       445 |
##                   |    32.710 |     9.004 |           |
## -------------------|-----------|-----------|-----------|
##      Column Total |       681 |      2474 |      3155 |
## -------------------|-----------|-----------|-----------|
##
##
```

This table is similar to what has been presented previously except that all proportions have been removed and only the counts and the so-called chi-square contribution are reported This is a 5-by-2 contingency table (the row and column totals are not considered part of the table). Assuming independence, the expected value E for each cell is

$$E = \frac{(\text{total of row}) \cdot (\text{total of column})}{\text{total count}}$$

For example, consider the cell "High School" and those who voted ($vote20 = 1$) which contains 1045 counts. The expected count under the null hypothesis that voting behavior is independent of education leads to the following:

$$E_{\text{High School},1} = \frac{1453 \cdot 2474}{3155} = 1139.373$$

Those calculations can be conducted for each cell. Then, the χ^2-test statistic is calculated as follows:

$$\chi^2 = \sum_{i=1}^{r \cdot c} \frac{(O_i - E_i)^2}{E_i}$$

where $r \cdot c$ is number of rows multiplied by the number of columns, O_i and E_i are the observed and expected count in a cell, respectively. The degrees of freedom are calculated as $(r-1) \cdot (c-1)$. For the example of respondents who finished high school (without seeking further degrees), the contribution to the χ^2 is calculated as follows:

$$\frac{(1054 - 1139.373)^2}{1139.373} = 6.397$$

Note that this is the value of the chi-square contribution reported in the table. Of course, instead of doing it manually, the following command can be used:

```
chisq.test(df$degree,df$schooling)
```

```
##
##  Pearson's Chi-squared test
##
## data:  df$degree and df$schooling
## X-squared = 12620, df = 16, p-value < 2.2e-16
```

Due to the p-value being below the significance level of 5%, we reject the hypothesis of independence. The χ^2 hypothesis test should only be used for qualitative data. For example, do not categorize income into quartiles to conduct a hypothesis test on whether voting depends on income.

11.2 Binomial Test

The binomial test is a hypothesis test about the probability of success. For example, consider the data `airlines` and the flights of Delta Airlines arriving in Indianapolis in 2023. Suppose that Delta claims that the chance of a flight arriving with a delay is less than 5%. A t-test on the proportion could be conducted if we had data about the individual flights. However, the data set only contains the aggregate numbers. In this case, we can use a binomial test.

Suppose that we pick December 2023 to analyze their claim. That month, 26 out of 370 flights arrived late which results in an average of 7.0% of flights being delayed. The binomial test can be implemented as follows:

```
binom.test(26,370,p=0.05,alternative=c("greater"))
```

```
##
##  Exact binomial test
##
```

```
## data:  26 and 370
## number of successes = 26, number of trials = 370, p-value = 0.05294
## alternative hypothesis: true probability of success is greater than 0.05
## 95 percent confidence interval:
##  0.0497021 1.0000000
## sample estimates:
## probability of success
##             0.07027027
```

The p-value is above the significance level of 5% and hence, we fail to reject the hypothesis.

11.3 Wilcoxon Signed-Rank Test

The Wilcoxon Signed-Rank Test is a non-parametric alternative to the paired t-test, used when the differences between paired observations cannot be assumed to follow a normal distribution. Instead of relying on the actual magnitudes of the differences, it considers their ranks, taking into account both the sign (positive or negative) and size of each difference. The procedure tests whether the median of these differences equals zero, which corresponds to testing whether two related samples come from the same population. Because it uses the ranks of the differences rather than their raw values, the Wilcoxon Signed-Rank Test is less sensitive to outliers and skewness, making it suitable for ordinal or non-normally distributed data. In the broader framework of hypothesis testing, it plays a role analogous to the paired t-test.

Suppose we have data on participants' productivity scores before and after a training program. The data for ten employees looks as follows:

```
before = c(72, 75, 70, 68, 73, 74, 69, 71, 70, 72)
after  = c(75, 78, 73, 70, 75, 77, 70, 73, 72, 75)
```

In a first step, we conduct a so-called Shapiro–Wilk test for normality to determine whether the difference is normally distributed. Note that the paired t-test assumes that the differences are normally distributed.

```
shapiro.test(after-before)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  after - before
## W = 0.78089, p-value = 0.008489
```

Since the data fails the test for normality, the Wilcoxon Signed-Rank test is preferred to the paired t-test.

```
t.test(after,before,paired=TRUE)
```

```
##
##  Paired t-test
##
## data:  after and before
## t = 10.854, df = 9, p-value = 1.801e-06
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
##  1.899818 2.900182
## sample estimates:
## mean difference
##             2.4
```

```
wilcox.test(after,before,paired=TRUE,exact=FALSE)
```

```
##
##  Wilcoxon signed rank test with continuity correction
##
## data:  after and before
## V = 55, p-value = 0.004995
## alternative hypothesis: true location shift is not equal to 0
```

11.4 Exercises

1. ***Chi-Squared Test*** (**): Use the `gss` data for this exercise. Select one of the following variables: *fulltime*, *government*, *married*, *gun*, or *deathpenalty*. Conduct a Chi-Squared hypothesis test between the variable chosen and education. Remember that the null hypothesis is independence between the variables. Report your results and interpret them as well.

2. ***Public Service Requests*** (**): A local government office claims that 80% of public service requests are successfully resolved within one week. You collect a random sample of 50 service requests, and find that 34 were resolved within one week. Formulate the null and alternative hypotheses to test the government's claim. Using R, perform the appropriate binomial

test to determine whether the true success rate differs from the claimed 80%. Report the test statistic, p-value, and interpret the result in context. Calculate and interpret the 95% confidence interval for the true success rate.

3. ***Waste Collection Program*** (***): A city implemented a new waste collection program and claims that the proportion of households that correctly separate recyclables has increased from 60% to 75%. You survey a random sample of 80 households, and find that 58 correctly separated recyclables.

    a. Clearly state the null and alternative hypotheses to test the city's claim.
    b. Perform a one-sided binomial test in R to determine whether there is evidence that the success rate is greater than 60%.
    c. Report the p-value and interpret it in context.
    d. Using the test's output, construct and interpret a 95% confidence interval for the true proportion.
    e. Suppose a larger city reports that 62 out of 100 households did so successfully. Use R to compare the two proportions and discuss whether both cities likely achieved the same success rate.